

HAMED DIAKITE

Software Engineer | Systems Developer | Networking & IoT

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PROFESSIONAL SUMMARY

Computer Science graduate with experience in system architecture, network infrastructure, and full-stack software development. Skilled in API design, distributed systems concepts, and IoT integration using Python, JavaScript, React, and Arduino. Experienced in designing scalable backend systems and configuring enterprise network environments.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, Swift, SQL

Frameworks: React, React Native, Flask, Express, TensorFlow.js

Architecture & Systems: System Architecture, API Design, Microservices, Distributed Systems, REST APIs

Networking: TCP/IP, VLANs, Routing Protocols (OSPF, RIP), Network Security, DNS, DHCP, NAT

Tools: Git, Cisco Packet Tracer, Xcode, IntelliJ, Visual Studio, PyCharm

Platforms: Linux, MacOS, Windows

Technologies: Arduino, IoT Systems, Bluetooth Protocols, NFC

ENGINEERING EXPERIENCE

IoT NFC Access Control System | *Mobile & Embedded Developer*

- Spearheaded a groundbreaking project to revolutionize access control through the development of cutting-edge NFC door-unlocking applications with seamless mobile-to-hardware integration.
- Designed system architecture integrating mobile applications, IoT hardware, and Bluetooth communication APIs for secure access control systems.
- Developed embedded IoT service communicating with mobile clients through API-driven Bluetooth protocols.
- Implemented security features including encrypted communication between mobile devices and Arduino controllers, ensuring data integrity, and preventing unauthorized access.
- Showcased technical expertise in cross-platform development and IoT integration, addressing the evolving landscape of secure access control systems.

TECHNOLOGIES USED: Swift, C++, Xcode, Arduino Uno, HC-06 (Bluetooth Module), NFC Protocol.

Network Architecture Design & Configuration

Network Systems Engineer | *Cisco Packet Tracer*

- Designed a simulated distributed systems network architecture including routers, VLAN segmentation, and routing protocols to support scalable infrastructure.
- Configured network topology including routers, switches, VLANs, and subnetting schemes to optimize traffic flow and network segmentation.
- Implemented security protocols including firewall rules, access control lists (ACLs), and network address translation (NAT) to protect network resources.
- Configured and troubleshooted network services including DHCP, DNS, and basic routing protocols (RIP, OSPF) in a virtualized environment.
- Documented network architecture, created network diagrams, and developed standard operating procedures for network maintenance and troubleshooting.

TECHNOLOGIES USED: Cisco Packet Tracer, TCP/IP, VLANs, Routing Protocols, Network Security, Linux Administration

WebGL 3D Graphics Rendering Engine

Graphics Systems Developer | *WebGL, Javascript*

- Developed an interactive 3D graphics application using WebGL and JavaScript, demonstrating proficiency in low-level graphics programming and shader development.
- Implemented custom vertex and fragment shaders using GLSL for realistic lighting models including Phong shading, ambient occlusion, and texture mapping.
- Created 3D object rendering pipeline with model transformation matrices, camera positioning, and perspective projection calculations.
- Designed and implemented user interaction controls for real-time camera manipulation, object rotation, and dynamic lighting adjustments using JavaScript event handlers.
- Optimized rendering performance through efficient buffer management and draw call batching, achieving 60 FPS on complex scenes.

TECHNOLOGIES USED: WebGL, GLSL, JavaScript, HTML5 Canvas, Linear Algebra, Computer Graphics Algorithms

AI-Powered Sign Language Translation System

Collaborative Software Engineering Project | *Team Developer (Agile)*

- Collaborated with a cross-functional team to develop an innovative sign language translation application aimed at improving accessibility and communication for the deaf and hard-of-hearing community.
- Contributed to the development of machine learning model integration for real-time gesture recognition and translation of American Sign Language (ASL) to text.
- Implemented front-end components for video capture, real-time processing display, and translation output using modern JavaScript frameworks.
- Participated in Agile development methodologies including sprint planning, daily standups, and code reviews to ensure high-quality deliverables.
- Conducted user testing sessions and incorporated feedback to improve accuracy and user experience of the translation interface.

TECHNOLOGIES USED: JavaScript, React, TensorFlow.js, Computer Vision APIs, HTML, CSS, Git

SOFTWARE ENGINEERING PROJECT

Electronic Health Records (EHR) System

Full-Stack Developer | *Python/Flask*

- Designed and implemented a secure healthcare data management system with RESTful APIs and role-based authentication, supporting scalable patient record access.
- Designed RESTful API architecture for secure healthcare data services, implementing scalable API design principles supporting distributed systems communication.
- Created a secure authentication system with role-based access control (RBAC) to ensure HIPAA-compliant data protection for patients, doctors, and administrative staff.
- Designed backend system architecture using Flask and SQLAlchemy ORM, enabling modular microservices-style backend components.
- Developed responsive front-end interface with dynamic forms for patient intake, medical record updates, and appointment scheduling.

TECHNOLOGIES USED: Python, Flask, SQLAlchemy, HTML, CSS, JavaScript, MySQL, RESTful APIs

Event Planner & Organizers Web Application

Front-End Developer | *React*

- Built a comprehensive event planning and management web application using React, enabling users to create, organize, and manage events with real-time updates.
- Implemented dynamic component architecture with React Hooks (useState, useEffect, useContext) for state management across multiple event planning features.
- Designed and developed responsive user interface using Tailwind CSS, ensuring seamless experience across desktop and mobile devices.
- Integrated real-time event calendar functionality with filtering and search capabilities, allowing users to discover and RSVP to events efficiently.
- Implemented user authentication and event creation workflows, including form validation, image uploads, and attendee management systems.

TECHNOLOGIES USED: React, JavaScript, Tailwind CSS, HTML, CSS, Firebase, RESTful APIs

WORK EXPERIENCE

GLOITR

Washington D.C | *June 2024 – August 2026*

Enterprise Network Engineer

- Installed and organized Main Distribution Frames (MDF) for enterprise network deployments.
- Installing network equipment – Access Points, routers, Switch UPS & PDUs.
- Installed network patching fiber jumpers/fiber patch cords and troubleshooting.
- Configured and labeled network switches during enterprise infrastructure deployments.

Harrisburg University of Science and Technology

Harrisburg, PA | *August 2023 – April 2024*

Connection Leader

- Orchestrated a series of engaging events, including tournaments, board games, movie nights, and virtual games, aimed at acclimating over 30 new students to the school system each semester.
- Initiated weekly check-ups on at least 30 students, ensuring a smooth transition and addressing individual concerns, resulting in a 25% increase in reported student satisfaction.
- Collaborated with 10+ student leaders across various departments to design and implement welcoming events, contributing to a 15% improvement in new student participation in school activities.

EXTRACURRICULAR & CERTIFICATION

Wells Fargo Personal Banker

February 2026

Personal Banker | Job Simulation on Forage

- Completed a Personal Banker Job Simulation involving customer service and financial guidance for a branch banking team.
- Practiced identifying customer needs and offering tailored support during service interactions.
- Strengthened communication skills through realistic client conversations and team collaboration.
- Applied problem-solving strategies to manage branch workflows and prioritize tasks in a fast-paced environment.

RELEVANT COURSEWORK

Computer Programming I & II | Information Technology | Algorithms | Data Structure | Linear Algebra | Operating System | Calculus I | Computer Organization and Architecture | Discrete Math I & II | Computer Networks | Computer Graphic | SQL Database | Advance Networking | Principle of Software Engineer & System Analysis

EDUCATION

Bachelor of Science – Computer and Information Science

Harrisburg University of Science and Technology

West Chester University of Pennsylvania

West Chester, PA

Masters - Computer Science

Anticipated Graduation: December 2028